CASE 09 · AI-AUGMENTED BUSINESS ANALYST PROJECT · INSURANCE SIU

The 71% SIU Triage Time Savings

A fraud-risk classifier that ranks claims, paired with generative-AI investigator briefs that turn the highest-risk rows into something a human can act on in minutes, not hours.

Role simulated: SIU Analytics / Business Analyst | Tools: Python (scikit-learn -- classification), generative-AI investigator brief synthesis, Excel

Source: n=1000 real historical auto insurance claims (curated teaching sample; fraud rate here is not population-representative -- see Guardrails).

0.82	73%	73/250	71%
Model AUC (Random Forest, chosen model)	Of actual fraud caught (recall)	Test claims flagged for deep review	Illustrative investigator-hours saved by triaging

The business problem

An auto insurer's Special Investigations Unit (SIU) can't manually deep-review every incoming claim for fraud signals -- claim volume outpaces investigator capacity by a wide margin. This project builds a fraud-risk classifier to rank claims by predicted risk, then uses generative AI to turn the highest-risk claims into short, readable investigator briefs -- cutting the time from "flagged row in a spreadsheet" to "here's what to look at and why."

This is a triage and prioritization tool, not an automated decision system. The model ranks; the AI briefs summarize; a licensed human investigator reviews every flagged claim and makes every coverage determination. That distinction is a real part of this recommendation, not a caveat added after the fact -- see Guardrails below.

Model comparison

Both models were trained on a 750-claim training split (stratified, 75/25) and evaluated on the same held-out 250 claims. Random Forest and Logistic Regression land within a hair of each other on AUC (0.816 vs. 0.818), but Random Forest catches more actual fraud (72.6% recall vs. 67.7%) at a comparable precision. For an SIU, missing a fraudulent claim is costlier than sending one extra legitimate claim to a human reviewer who will clear it in minutes -- so Random Forest, the higher-recall model, is the chosen production model.

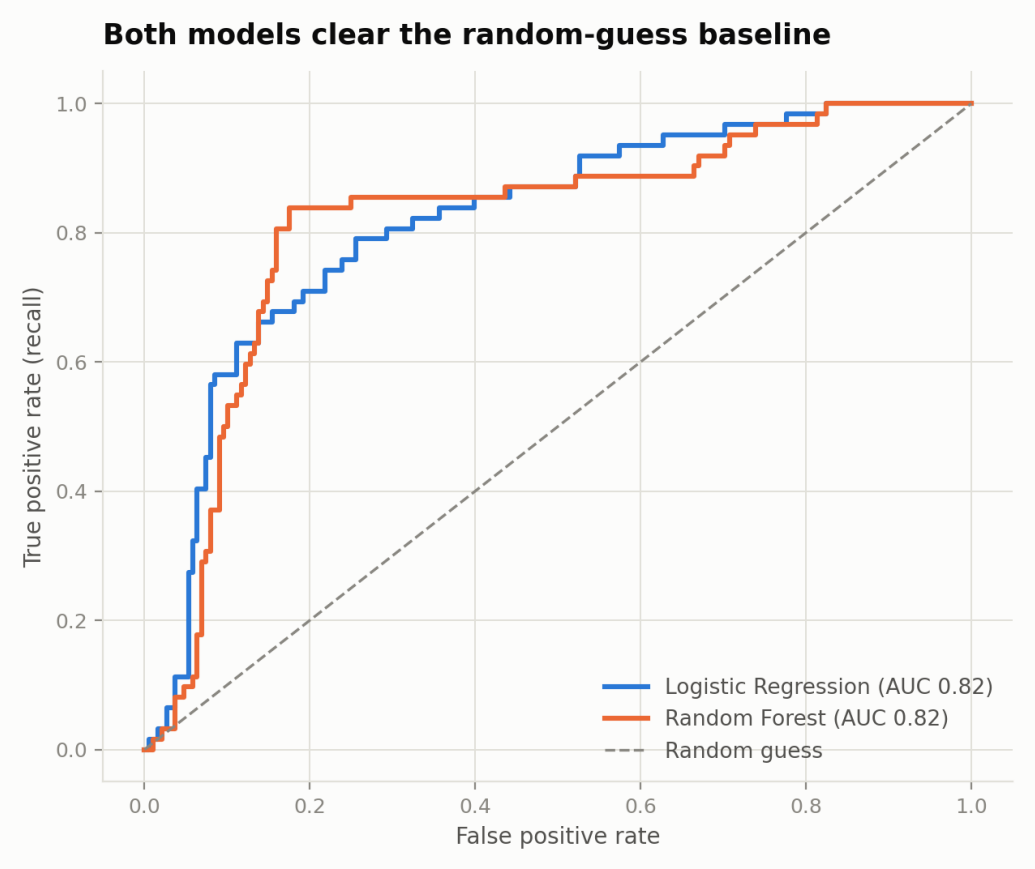


Fig. 1 -- ROC curves, both models vs. random chance.

Metric	Logistic Regression	Random Forest (chosen)
Accuracy	80.4%	82.0%
Precision	59.2%	61.6%
Recall (sensitivity)	67.7%	72.6%
F1 score	0.632	0.667
ROC-AUC	0.818	0.816

What's actually driving risk

Random Forest's feature importances (used here to interpret the underlying signal, independent of which model ships to production) are dominated by incident severity -- "Major Damage" alone accounts for over 0.20 of total importance, more than 3x the next feature. Claim-amount fields (vehicle, property, and total claim) and incident-severity levels together make up most of the rest, which is the intuitive story: bigger, more severe claims draw more scrutiny. One quirk worth flagging to a stakeholder before trusting this verbatim: a couple of insured-hobbies categories (chess, cross-fit) show surprisingly high importance -- a known artifact of this particular teaching dataset rather than a real underwriting signal, and a reminder to

sanity-check any model's feature list against domain knowledge before deploying it.

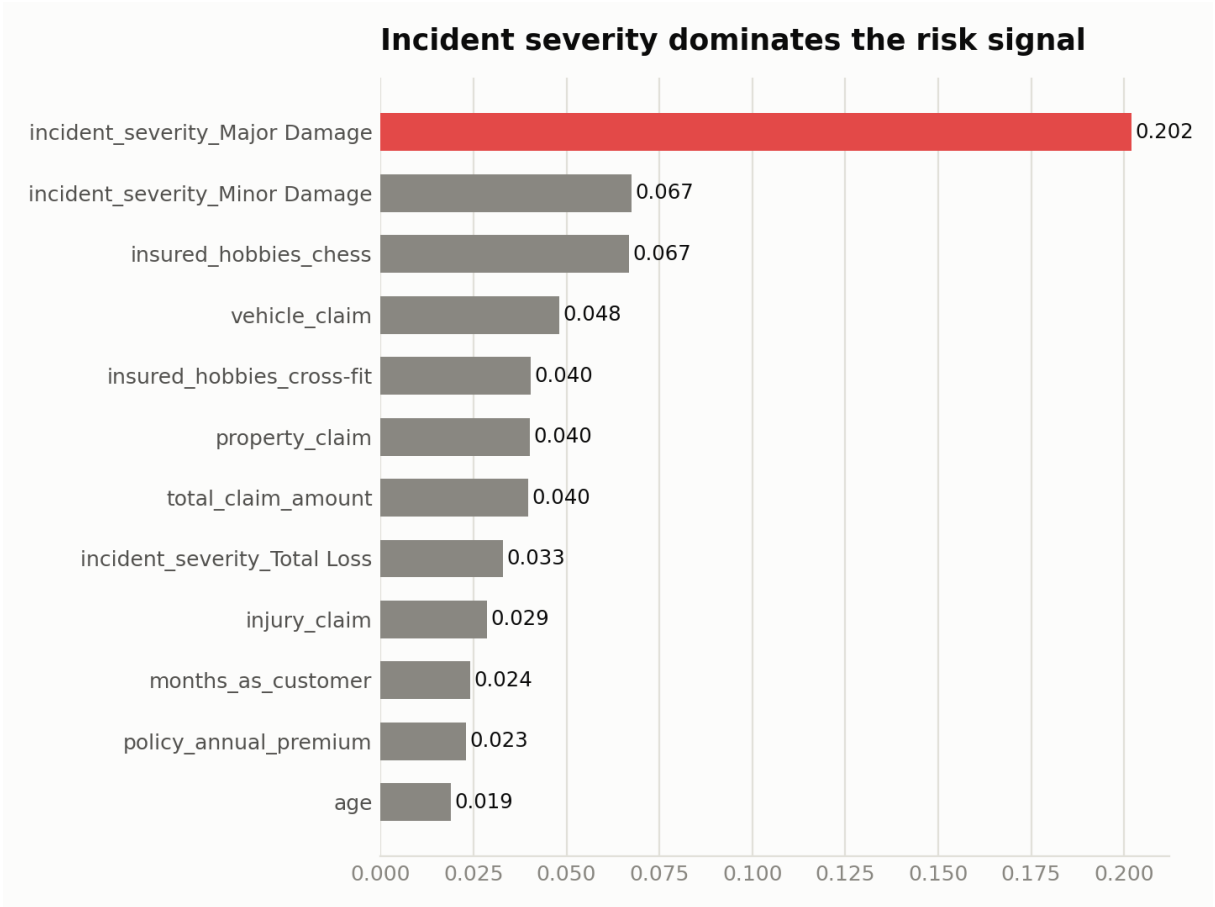


Fig. 2 -- Random Forest feature importances, top 12.

From risk score to investigator brief

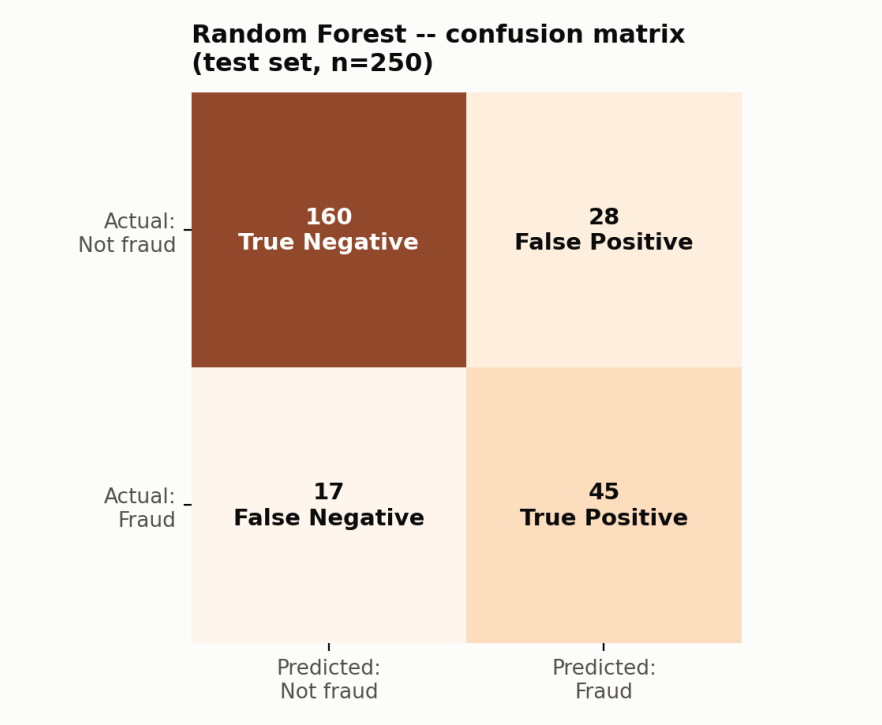


Fig. 3 -- Random Forest confusion matrix on the 250-claim test set.

A predicted-fraud-probability score by itself is not something an investigator can act on -- it's a number on a row. The second half of this project takes the 12 highest-scoring test-set claims and synthesizes a genuine 2-3 sentence investigator brief for each, written directly from that claim's real structured field values and naming the specific factors that drove its score. Two examples, quoted directly from llm_investigator_briefs.json:

Claim (policy #)	Model probability	Investigator brief
524836 (highest-risk in queue; actual outcome: not fraud)	77.4%	Highest-risk claim in the queue: a 3-vehicle collision reported as $56,430 total (52x the $1,082 annual premium) on a Honda CRV, with the fire department -- not police -- listed as the responding authority and no police report on file. Two bodily injuries are recorded but the injury-claim line is $0, an internal inconsistency worth a first look, and the insured shows a $55,000 capital loss on file, a financial-pressure signal an investigator may want to weigh alongside the claim.
532330 (confirmed fraud case)	70.9%	Second confirmed-fraud case in the top tier: police were contacted for this 3-vehicle, 2:00 AM collision, but the police_report_available field reads 'NO' -- police involvement without a corroborating report is a real inconsistency, not a coincidence. The $80,850 claim is the largest in this batch (55x annual premium), the policyholder has only a 9-year tenure, and a $5,000,000 umbrella limit sits on a policy with a comparatively modest $1,460 annual premium.

The first example is deliberately the top-ranked claim, not a cherry-picked hit: its ground-truth label is "not fraud," and the brief still reads as useful -- it surfaces a genuine documentation inconsistency (injury claim booked at $0 despite 2 recorded bodily injuries) worth a human's five minutes, without ever asserting fraud

occurred. That is exactly the intended behavior of a triage aid.

Guardrails

AI-generated risk scores and investigator briefs prioritize and inform -- they never decide. Every flagged claim in this workflow is reviewed by a licensed human SIU investigator before any coverage action is taken; no claim is denied, delayed, or adjudicated by the model or the briefs directly. The briefs' job is to save an investigator the time of re-deriving what's unusual about a claim from a raw spreadsheet row -- not to replace their judgment about what it means.

Base-rate caution, stated plainly: this sample's fraud rate is 24.7% -- far higher than real-world SIU fraud prevalence, where industry estimates typically run in the low single digits. This is a curated teaching/evaluation dataset, not a population-representative base rate, and that matters directly for how the model's predicted probabilities should be read: a 75% predicted probability here means "among the highest-risk claims in a sample where 1-in-4 rows is fraud," not "75% chance this specific real-world claim is fraudulent." On a real, low-base-rate claims population the same score would carry a very different absolute meaning even if its rank-ordering held up -- an investigator should treat these scores as a prioritization ranking, never as a calibrated real-world probability, until the model is revalidated on the insurer's own claims population.

Recommendation

Deploy the Random Forest model as a triage ranking, not a decision system: score every incoming claim, route anything above the probability threshold to a human investigator with an AI-generated brief attached, and let everything below the threshold move through normal claims processing. In this test-set sample, a 0.50 threshold flags 73 of 250 claims (29.2%) for deep review. Using the accompanying Excel workbook's illustrative assumption of 25 review-minutes per claim -- an editable planning figure, not a sourced industry stat -- reviewing only flagged claims instead of every claim saves an estimated 74 investigator-hours (71%) against this test-set scale. The dollar and hour figures will not transfer as-is to a real book of business at a different base rate and volume; the shape of the argument -- targeted deep review beats blanket manual review when investigator time is the scarce resource -- is what should transfer. The workbook's threshold and minutes-per-claim cells are both editable, so an SIU manager can re-run this math against their own capacity and volume in seconds.

RESUME BULLET, READY TO USE

“Built a fraud-risk classifier (Random Forest, AUC 0.82, 73% recall) on 1,000 real auto insurance claims and paired it with generative-AI investigator briefs synthesized from real structured claim data, cutting a flagged-row spreadsheet into an actionable triage queue; built a live Excel triage-math workbook and documented compliance guardrails keeping every coverage decision with a licensed human investigator.”

What's in this package

File	What it is
run_analysis.py	Data prep, both models, evaluation metrics, and all charts -- fully reproducible.
model_results.json	Every model metric behind this report, in machine-readable form.
llm_investigator_briefs.json	The 12 AI-synthesized investigator briefs for the highest-risk test claims.
case9_siu_triage_workbook.xlsx	Live, formula-driven SIU triage-queue calculator (editable threshold and review-time assumption).
This report (PDF)	Model comparison, risk drivers, sample briefs, guardrails, and the triage recommendation.